

Designing Digital Health Tools to Address Stigma and Structural Barriers across the Mental Health Care Ecosystem

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Abstract

This doctoral research examines how digital health technologies can address structural barriers and pervasive stigma across the mental health care ecosystem by studying three distinct yet connected populations: individuals with obesity navigating weight-stigmatizing healthcare encounters, mental health therapists experiencing burnout and compassion fatigue, and college-going young adults in underserved contexts facing barriers to mental wellbeing resources. This work is structured into four studies: the first two focusing on needs assessment and design of an educational tool to reframe narrative on obesity, followed by design and evaluation of an LLM-powered intervention to support mental wellbeing of therapists, and the final study looking at psychotherapeutic frameworks and design features that make chatbot based wellbeing tool engaging for young adults. By using qualitative and participatory research methods, this research contributes to HCI's understanding of designing equitable mental health technologies that address systemic barriers preventing marginalized populations from accessing the care they need.

CCS Concepts

• Human-centered computing → Empirical studies in HCI; • Applied computing → Health informatics.

Keywords

Digital health interventions, mental health, structural barriers, weight stigma, therapists, young adults, co-design, LLM-powered interventions

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1 Introduction

Access to mental health care in the United States is marked by significant disparities. Although in recent years there has been a growing recognition of mental health as essential to the overall wellbeing of an individual, systemic barriers and pervasive societal stigma

prevent millions from accessing quality care [17]. These structural challenges manifest differently across populations. Individuals with obesity are subject to weight stigma throughout their lives and especially in clinical encounters that not only impact their physical health outcomes but also exacerbate their mental health through internalized blame and shame, and care avoidance [2]. Mental health therapists experiencing compassion fatigue and burnout are at risk of compromised mental wellbeing themselves due to the emotional weight of their work, overwhelming caseloads, particularly after COVID-19, and pervasive shame within their professional circle around admitting stress, along with limited institutional support systems to navigate these challenges [1, 3]. Similarly, college students avoid seeking mental health care due to limited knowledge of available resources, fear of privacy being compromised by involving authorities, and persistent stigma around mental wellness that makes formal seeking of help daunting for young adults [11, 14]. While these populations appear distinct, they are connected by shared systemic failures that portray mental health challenges as personal failings at the level of individual responsibility rather than as structural conditions and circumstances that can be addressed through proactive, compassionate care and support. My dissertation, therefore, treats these cases as comparative studies for understanding how stigma and structural barriers manifest differently within a common mental health care ecosystem.

My doctoral research examines how digital mental health technologies can address stigma and systemic barriers broadly across the mental health care system by studying different vulnerable populations. By taking a socio-ecological perspective on vulnerability [18] to unpack how stigma, barriers to access, and inadequate support tools create unifying challenges despite disparate contexts. Through my thesis work, I aim to design and evaluate health interventions for three use cases: a patient-facing educational tool that reframes stigmatizing narratives and supports self-advocacy, a provider-companion support system that addresses therapist burnout and inadvertently enhances care quality, and an AI-powered, personalized wellbeing tool for on-demand support. This multi-stakeholder research approach recognizes that adequate and sustainable mental health care requires interventions that are not just patient or client-facing but also cater to the needs of providers who care for them, and propose changes to the system that enable timely support and intervention.

2 Related Work

The design of digital mental health support tools is well researched in HCI for various populations, including youth and individuals



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with stigmatized health conditions [5, 10, 18]. However, the intersection of structural barriers in the design of digital health interventions remains an underexplored area of research.

Weight stigma is deeply rooted in societal beliefs that attribute excess weight to individual failure, leading to perceptions of people with obesity as lazy, undisciplined, and lacking willpower [13]. The HCI research has primarily examined weight-related issues through the lens of disclosure on social media or in online support communities, and of behavior change technologies [2, 8]. Recent scholarship has explicitly focused on fat studies perspectives to draw attention towards weight bias in digital systems and propose design justice systems that challenge normative body assumptions. However, a critical gap remains in designing tools that address stigma in healthcare encounters where it most directly harms not just physical but also mental health outcomes [7, 16].

Additionally, the HCI community has extensively examined compassion fatigue and emotional labor across different care contexts, including caregivers of individuals with serious chronic conditions, healthcare workers during crises, and community health workers in resource-constrained settings [3]. While significant, this body of work largely focuses on professional outcomes, including care delivery quality, retention, and effectiveness, rather than care providers' own mental wellbeing. Specifically, Simon et al have designed tools to support therapists' work through effective documentation systems and clinical decision support [15]. This represents a critical gap in understanding therapist wellbeing as their profession accompanies the emotional toll of clients' trauma, managing overwhelming caseloads in burdened systems and professional norms that are not encouraging towards expressing distress or seeking help [9].

Although there is substantial work on mental health technologies for youth, including mood tracking apps [4], peer-support tools, and AI-powered interventions, low engagement with these tools remains a challenge [6, 12]. This gap becomes even more important to overcome in rural settings, where resources are already limited and digital wellbeing tools might be the only option available.

3 Research Objectives

The goal of my PhD thesis is to understand how digital technologies can be designed to help mitigate stigma-related and structural barriers that restrict access to quality care for different marginalized populations. Specifically, it aims to understand how pervasive societal stigma towards certain populations reduces the quality of care being given and simultaneously exacerbates the mental wellbeing of affected individuals. The next step in this study explores how stigma can be reframed through technology, enabling affected individuals to better advocate for themselves. Additionally, it also explores the design opportunities to better support professional therapists who commonly experience compassion fatigue and burnout. Finally, it aims to catalogue design features that make emerging technologies sufficient and engaging tools to support overall mental wellbeing. Overall, these objectives aim to generate transferable design knowledge for equitable digital mental health systems.

4 Research Approach

For my dissertation, I use human-centered design (HC) approaches throughout the design, study, and evaluation of technologies in close collaboration with the populations I am to support through my work. My collaborators include patients seeking care for weight concerns, mental health providers at risk of burnout and compassion fatigue, and college-going young adults in rural settings. Additionally, I work closely with clinicians, medical students in training, and psychologists. Given that my work addresses sensitive topics marked by stigma, I primarily employ qualitative research methods, including semi-structured interviews, co-design sessions, and focus groups, and then conduct thematic analysis to identify patterns while preserving individual experiences. Ethical practices are essential for research of this nature. To ensure participant safety and wellbeing, I take informed consent via phone call before the study and only collect minimal personal data. Additionally, I remain attentive to how my positionality shapes research interactions. I match interviewer-participant demographics where possible (e.g., female researchers interviewing female participants about body-related experiences). I work closely with MD-PhD students and clinicians to validate that medical information is scientifically sound and pitched at appropriate health literacy levels, specifically for the weight stigma project. I also actively participate in workshops where experts from different disciplines collectively narrow the project's focus and assess clinical feasibility, ensuring our approach is grounded in participants' needs.

5 Research to Date

5.1 Study One: Learning about the lived experiences of weight stigma in healthcare encounters

I conducted a formative needfinding study to understand the experiences of seeking care for weight or obesity, using semi-structured qualitative interviews. This study focused on learning broadly about care experiences, with the main aim of examining how weight stigma manifests in healthcare encounters.

These in-depth interviews began with general questions about healthcare experiences, including the frequency of visits, facilitators, and barriers to seeking care, and then proceeded to questions specific to weight-related healthcare appointments. Weight stigma has been shown to be one of the main factors for care avoidance, so questions towards the end also focused on experiences of seeking care for preventative healthcare screenings, such as for breast and cervical cancer. We interviewed 14 participants for this study, and the majority held a unique position in the healthcare field, working in various capacities while also being patients seeking care for weight concerns.

We conducted thematic analysis of the interview data and found deep-rooted weight stigma in healthcare, as indicated by a lack of appropriate equipment for larger bodies and a disproportionate emphasis by providers on weight for unrelated health concerns and being recommended generic food and exercise plans for weight loss despite the shown efficacy of adopting a personalized approach to care for weight. These interviews also uncovered the emotional toll of weight-based discrimination, such as individuals indulging in

coping mechanisms that they know are not beneficial for them and blaming themselves for their health issues. Some participants also shared how they avoid hospital visits due to anticipated shame or comments from staff. While sharing the positive aspects of their appointments and reflecting on what made them feel seen and heard, the participants noted that their healthcare providers used visual aids to thoroughly explain the bodily processes linked to weight gain and slow weight loss, and answered their questions with descriptions and mechanisms.

This study served as the basis for the next study, which aimed to address knowledge gaps regarding the causes of excess weight and its links to co-occurring health conditions. This led to the co-design of an educational tool to better understand the causes of obesity.

5.2 Study Two: Co-design of an educational tool to reframe obesity

After identifying clear knowledge gaps in understanding obesity, this study aimed to address persistent weight stigma in healthcare by collaboratively designing an educational portal that reframes obesity away from individual blame and toward a person-centered, medically grounded understanding. The insights from study 1 informed our iterative co-design sessions and content reviews, where the lived experiences of individuals seeking care for weight concerns and clinicians' practical constraints were treated as design resources in parallel. Throughout these sessions, we had to navigate tensions between clinical framings of obesity and perspectives from fat liberation [16], aiming to preserve medical credibility while avoiding stigmatizing language and assumptions. The resulting patient-facing portal integrated narrative accounts, generously incorporated infographics and figures for easier explanation, and actionable guidance that individuals with obesity can use to better understand obesity's multifactorial causes, treatment options, and mental health implications, to advocate for a more personalized, comprehensive care.

This study also narrowed the design requirements for weight-inclusive educational tools to include person-first framing, the use of accessible visual aids to break down complex concepts, the inclusion of lived experience as authoritative knowledge, and content that encourages more open patient-provider dialogue. Overall, this study was pivotal in demonstrating how participatory design can produce educational interventions that are both clinically sound and socially grounded, offering a scalable model for addressing stigma in other health contexts.

6 Ongoing and Proposed Research

The first two studies, although focused on weight stigma as a use case, introduced me to HCI research with marginalized populations and revealed promising research directions for future work. While subsequent studies appear distinct in their themes, they are connected by a common thread: addressing structural barriers to mental health resources across different populations. As such, I want to understand how digital support tools can be designed for other communities that similarly lack ready access to mental health resources, including mental health providers themselves and young adults in underserved contexts.

6.1 Study Three: Design and implementation of LLM-powered support tool for mental health providers

The preliminary study for this work has been accepted at CSCW 2024, where semi-structured interviews with 8 mental health workers revealed a high risk of burnout, secondary traumatic stress, and compassion fatigue [3]. Although there were mixed views on using technology for mental health support, some participants were open to adopting novel approaches for self care. Given the limited sample size, I plan to expand the formative needs assessment by examining experiences shared by therapists anonymously on dedicated platforms such as subreddits and Facebook groups. Beyond needs assessment, this analysis will also help gauge how open this population is towards technology adoption and sustained use.

Specifically, I want to answer the following two research questions with this study: (1) How do therapists experience and make sense of emotional distress in their therapeutic work, and what contextual factors (e.g., client interactions, systemic factors, and relational dynamics) shape these experiences? (2) Can we build a system that reliably detects emotionally significant or triggering moments during therapy sessions and deliver timely, context-aware support to therapists immediately after the session ends?

In answering the first question, I hope to develop design principles for a support system that detects potentially triggering moments based on both session content and individual therapists' histories. My vision is to create an intelligent, generative AI system that reframes therapy session content in ways that are reaffirming for the therapist. I believe this connects well to my previous work on shifting narrative from an obvious, stigmatizing perspective (as with pervasive weight bias in obesity care) to a more empathetic, grounded framing that supports provider wellbeing.

The main challenges in this study center on privacy concerns about AI listening to therapy sessions, the fear of over-reliance on AI models for provider support, and hesitation among the target population to adopt additional technology in their practice. We already have established contacts from the earlier study, and I will leverage that to closely work with providers in mental health services throughout the design process and co-design the intervention at every stage. This participatory approach will help providers understand how AI systems can be privacy-preserving while remaining cognizant of their needs. At the same time, I will also remain open to adapting my approach if strong resistance to technology use persists, potentially pivoting to alternative support modalities that better align with providers' preferences and work context.

6.2 Study Four: Therapeutic foundations, design practices, and engagement in mental wellbeing interventions for young adults

The focus of this study is to ensure that digital support tools are grounded in evidence-based psychotherapy models and employ appropriate design principles to be engaging. This is important in the context of young adults living in rural areas where access to in-person services is limited, and hence it is critical to ensure that technologies for this population are not just based on established therapeutic frameworks but attract sustained use to be beneficial,

as a high abandonment rate remains a big issue for digital mental health interventions [11]. Both of these challenges mirror the structural barriers I have observed throughout my dissertation work, and this final study will help me strengthen my thesis.

This study will be based on a systematic review to address engagement as a central focus through the following research questions: (1) To what extent and in what ways have evidence-based psychotherapy principles been translated into design features and interaction patterns, and how do these choices relate to engagement outcomes? (2) In what ways have mental health and behavior change chatbots been framed, in terms of their styles, character presentation, and intent to encourage sustained engagement among young adults? By systematically examining the literature, I aim to map how therapeutic foundations shape engagement patterns, identify design practices associated with sustained use, and develop evidence-based recommendations for designing chatbot-based interventions that young adults continue to use beyond initial novelty.

As part of the outcomes for this study, I will develop a framework that captures therapeutic models, design features, and, specifically, engagement-related metrics, including interaction patterns, dropout timing, user versus system initiation, chatbot characteristics, and reported other potential barriers to continued use. I envision this framework enabling the identification of design principles that overcome engagement challenges, paralleling how my educational tool on obesity addresses clinical stigma barriers and my idea for a therapist tool addresses institutional support barriers. The findings from this final study will contribute to the development of an intervention for young adults in rural New England and the broader understanding of how digital mental health tools can sustain engagement by addressing the structural barriers that prevent young adults from accessing support.

7 Conclusion

My research focuses on showcasing how digital health technologies can be designed to tackle not just individual mental health challenges, but the structural barriers and stigma that prevent marginalized populations from accessing care. Through studies on experiences of weight stigma in clinical encounters, providers in therapy services experiencing burnout and compassion fatigue, and youth mental wellbeing, I demonstrate that sustainable mental health support requires a nuanced approach which empowers individuals to advocate for themselves in the face of pervasive societal stigma, supports providers whose wellbeing is essential to care delivery, and designs wellbeing interventions that maintain engagement by operationalizing foundational therapeutic frameworks and design principles rather than simply hoping individuals would adopt new technologies. By focusing on these contributions, my research strives to empower marginalized populations and enhance their interaction with digital tools in meaningful and equitable ways.

References

- [1] Richard E Adams, Joseph A Boscarino, and Charles R Figley. 2006. Compassion fatigue and psychological distress among social workers: A validation study. *American Journal of orthopsychiatry* 76, 1 (2006), 103–108.
- [2] Stevie Chancellor, Andrea Hu, and Munmun De Choudhury. 2018. Norms Matter: Contrasting Social Support Around Behavior Change in Online Weight Loss Communities. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. ACM, New York, NY, USA, 1–14. doi:10.1145/3173574.3174240
- [3] Aishwarya Chandrasekaran, Rebecca Currano, Vafa Batool, Kaiping Chen, Elizabeth L Murnane, David Sirkin, and Matthew Louis Mauriello. 2024. Therapy for therapists: Design opportunities to support the psychological well-being of mental health workers. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW2 (2024), 1–34.
- [4] Leigh R. Cowan, Kavous Salehzadeh Niksirat, Alessandro Silacci, Daniele Quercia, Daniel A. Epstein, and Christian Holz. 2022. Understanding the Why of Reflection for Self-Tracking During a Global Pandemic. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM, 1–14. doi:10.1145/3491102.3517641
- [5] Claudia Daudén Roquet, Nikki Theofanopoulou, Jaimie L Freeman, Jessica Schleider, James J Gross, Katie Davis, Ellen Townsend, and Petr Slovak. 2022. Exploring situated & embodied support for youth's mental health: design opportunities for interactive tangible device. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–16.
- [6] Jacinta Jardine, Camille Nadal, Sarah Robinson, Angel Enrique, Marcus Hanratty, and Gavin Doherty. 2024. Between rhetoric and reality: real-world barriers to uptake and early engagement in digital mental health interventions. *ACM Transactions on Computer-Human Interaction* 31, 2 (2024), 1–59.
- [7] Rebecca M Jonas, Ankolika De, and Kelley Cotter. 2024. "We happen to be different and different is not bad": Designing for Intersectional Fat-Positive Information-Seeking. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [8] Nadia Karizat and Nazanin Andalibi. 2023. "I like to See the Ups and Downs of My Own Journey": Motivations for and Impacts of Returning to Past Content About Weight Related Journeys on Social Media. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW1 (2023), 1–41.
- [9] P Alex Linley and Stephen Joseph. 2007. Therapy work and therapists' positive and negative well-being. *Journal of Social and clinical Psychology* 26, 3 (2007), 385–403.
- [10] Juan F Maestre, Daria V Groves, Megan Furness, and Patrick C Shih. 2023. "It's like With the Pregnancy Tests": Co-design of Speculative Technology for Public HIV-related Stigma and its Implications for Social Media. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–21.
- [11] Patrick D McGorry, Cristina Mei, Andrew Chanen, Craig Hodges, Mario Alvarez-Jimenez, and Eóin Killackey. 2022. Designing and scaling up integrated youth mental health care. *World Psychiatry* 21, 1 (2022), 61–76.
- [12] Bruna Oewel. 2025. Designing for Engagement in Digital Mental Health Services. In *Companion Publication of the 2025 ACM Designing Interactive Systems Conference*. 131–136.
- [13] Rebecca M Puhl, Mary S Himmelstein, and Rebecca L Pearl. 2020. Weight stigma as a psychosocial contributor to obesity. *American Psychologist* 75, 2 (2020), 274.
- [14] Sang-Wha Sien, Shalini Mohan, and Joanna McGrenere. 2022. Exploring design opportunities for supporting mental wellbeing among East Asian university students in Canada. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [15] Witold Simon, Michael J Lambert, Mitchell W Harris, Gregory Busath, and Aaron Vazquez. 2012. Providing patient progress information and clinical support tools to therapists: Effects on patients at risk of treatment failure. *Psychotherapy research* 22, 6 (2012), 638–647.
- [16] Aisha Sobey. 2024. Conceptualising fatness within HCI: A call for fat liberation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [17] Substance Abuse and Mental Health Services Administration. 2023. *Key Substance Use and Mental Health Indicators in the United States: Results from the 2022 National Survey on Drug Use and Health*. HHS Publication PEP23-07-01-006. Center for Behavioral Health Statistics and Quality, Substance Abuse and Mental Health Services Administration, Rockville, MD. <https://www.samhsa.gov/data/report/2022-nsduh-annual-national-report>
- [18] Xinru Tang, Gabriel Lima, Li Jiang, Lucy Simko, and Yixin Zou. 2025. Beyond "Vulnerable Populations": A Unified Understanding of Vulnerability From A Socio-Ecological Perspective. *Proceedings of the ACM on Human-Computer Interaction* 9, 2 (2025), 1–30.